

EAAP: Enterprise AI Architecture Pattern

A Governance Architecture for Deterministic Enterprise AI Systems

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Abstract

Enterprise adoption of artificial intelligence has accelerated faster than the development of corresponding governance architecture. While large language models and multi-agent systems demonstrate increasingly capable reasoning behaviour, most enterprise deployments remain architecturally incomplete: governance is frequently implemented as policy rather than structural control, evidence provenance is often absent, confidence is rarely calibrated in operationally meaningful ways, and AI systems are commonly integrated directly into operational environments without execution boundaries.

This paper presents the Enterprise AI Architecture Pattern (EAAP), a governance-oriented architectural model for enterprise AI systems operating in high-trust and high-impact environments. EAAP defines a layered orchestration and governance framework intended to constrain, structure, and audit AI-assisted operational reasoning. The framework introduces governance-deterministic orchestration principles, domain-isolated reasoning agents, evidence classification models, calibrated confidence gates, human approval boundaries, and execution isolation through controlled automation gateways.

Unlike model-centric approaches that treat the underlying large language model as the system itself, EAAP positions the model as a replaceable reasoning component operating within a larger governed architecture. The framework is designed to support air-gapped and sovereign AI deployments, including enterprise environments where external API connectivity is prohibited or operationally unacceptable.

This paper defines the architectural principles, threat model, orchestration model, governance layers, and operational constraints of EAAP, presents an illustrative operational scenario, and identifies the current limitations and future research directions of the framework.

Keywords: enterprise AI governance, orchestration, governance determinism, sovereign AI, multi-agent systems, evidence classification, auditability, AI architecture, air-gapped AI, threat model, operational governance

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1. Introduction

1.1 Problem Statement

Enterprise deployment of artificial intelligence has accelerated rapidly across infrastructure operations, software engineering, cybersecurity, customer support, compliance analysis, and enterprise knowledge management. However, the governance architecture surrounding these deployments has not matured at the same rate.

Most current enterprise AI implementations exhibit one or more recurring structural weaknesses:

- AI recommendations are generated without evidence provenance.
- Confidence levels are either absent or operationally meaningless.
- Human approval processes are advisory rather than structurally enforced.
- AI systems are granted direct operational access to infrastructure.
- Multi-domain reasoning occurs without orchestration or arbitration.
- Governance controls are implemented as organisational policy rather than architectural constraint.

These weaknesses create a governance gap between AI capability and operational trustworthiness. The problem is not primarily model intelligence. Increasing model capability alone does not resolve the absence of auditability, controlled execution, operational traceability, or governance enforcement. This paper argues that enterprise AI requires governance architecture, not merely more capable models.

The Enterprise AI Architecture Pattern (EAAP) is proposed as a governance-oriented architectural framework intended to constrain and structure AI-assisted operational reasoning in enterprise environments. EAAP is not a model, product, or software platform. It is an architectural control pattern.

EAAP is a conceptual governance architecture and not a certification framework.

1.2 Authorship and Intent

This paper presents a governance-oriented enterprise AI architecture framework intended to increase operational safety, auditability, and governance enforcement in enterprise AI systems. The framework constrains AI behaviour through structural mechanisms rather than relying solely on organisational policy or operator discretion.

Nothing in this paper should be interpreted as advocating autonomous operational decision-making without human oversight. The framework is explicitly designed around the principle that AI systems provide structured recommendations while accountability and operational authority remain human responsibilities.

1.3 Research Contributions

This paper makes the following specific contributions:

- A governance-oriented layered architecture pattern for enterprise AI systems operating in high-trust environments, positioning the large language model as a replaceable reasoning component.

- A formal definition of governance determinism as a new architectural concept distinct from computational determinism.
 - A threat model identifying nine failure modes that governance-oriented AI architecture must address.
 - A formal evidence classification model requiring AI outputs to disclose the provenance and basis of all recommendations.
 - A tiered human approval boundary model that structurally enforces human oversight for high-impact operations.
 - An execution isolation model prohibiting direct AI access to operational infrastructure through a controlled Automation Gateway.
 - A Conditional Override mechanism that replaces false certainty with escalation when confidence or evidence is insufficient.
 - An architectural model explicitly designed to support air-gapped and sovereign deployment environments.
 - An illustrative operational scenario demonstrating the EAAP governance lifecycle across a realistic Tier 2 enterprise infrastructure action.
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2. Background and Related Work

2.1 Enterprise AI Adoption

Enterprise AI adoption has expanded rapidly following the emergence of large language models capable of generalised reasoning and natural language interaction. Organisations increasingly use AI systems for operational analysis, infrastructure planning, policy interpretation, cybersecurity support, automation assistance, software generation, and enterprise search.

At the same time, many organisations have experienced the emergence of uncontrolled "Shadow AI" usage, where employees provide proprietary information to externally hosted AI services without governance controls. This creates substantial compliance concerns in regulated industries including defence, healthcare, critical infrastructure, financial services, and government operations.

2.2 Existing AI Governance Approaches

Current AI governance approaches are typically organised around ethical principles, organisational policy, compliance frameworks, or model alignment. The NIST AI Risk Management Framework [2] provides a risk-oriented voluntary framework structured around four functions: Govern, Map, Measure, and Manage. ISO/IEC 42001:2023 [3] defines management system requirements for AI, establishing organisational accountability and control mechanisms. The EU AI Act [6] introduces risk-based regulatory requirements for AI systems deployed within EU jurisdictions.

Floridi et al. [1] proposed the AI4People ethical framework, establishing principles of beneficence, non-maleficence, autonomy, justice, and explicability. However, many governance approaches remain insufficiently architectural. Governance is often expressed as policy language or high-level principles without defining the operational control structures required to enforce those principles structurally. EAAP addresses this gap.

2.3 Multi-Agent Systems and Orchestration Frameworks

Recent progress in LLM-based multi-agent systems has produced several relevant frameworks. Wu et al. [9] introduced AutoGen, a framework enabling multi-agent conversations for task completion through specialised agent roles and structured message passing. Hong et al. [10] proposed MetaGPT, a meta-programming framework assigning human-like roles to agents in a collaborative software engineering workflow.

Lewis et al. [11] introduced Retrieval-Augmented Generation (RAG), enabling language models to retrieve from external knowledge stores before generating responses. This mechanism underpins the EAAP Knowledge and Evidence Layer. Zhao et al. [24] provide a comprehensive survey of large language model architectures, noting that capability increases do not inherently produce governance improvements, directly motivating the EAAP governance-first principle.

Despite the practical value of these frameworks, none addresses enterprise governance as a primary architectural constraint. AutoGen and MetaGPT optimise for task completion. EAAP occupies a different design space: it optimises for governed, auditable, and human-overseen operation. These goals are complementary but architecturally distinct.

2.4 AI Safety and Human Oversight

Amodei et al. [15] identified concrete problems in AI safety including reward hacking, safe exploration, and scalable oversight. Hendrycks et al. [17] extended this work, cataloguing unsolved problems in ML safety including distributional shift and monitoring challenges in deployed systems.

Ouyang et al. [12] demonstrated that reinforcement learning from human feedback can align model outputs with human preferences. While EAAP does not prescribe training methodology, the principle that human preferences should structurally constrain AI behaviour is foundational to the approval boundary model.

Amershi et al. [14] identified 18 design guidelines for human-AI interaction systems, including making AI uncertainty visible and supporting efficient correction. EAAP's Conditional Override mechanism implements several of these guidelines as architectural constraints. Seshia et al. [18] proposed formal specification approaches for deep neural networks, motivating the principle that governance should be expressed as structural constraints rather than policy language. Doshi-Velez and Kim [16] established that operationally meaningful uncertainty quantification requires more than probability scores, informing EAAP's evidence classification model.

2.5 The Governance Gap

The central governance gap addressed by EAAP is the difference between declaring governance requirements and implementing governance structurally. Table 1 illustrates common failure modes.

Governance Requirement	Common Failure Mode
Human approval required	Approval can be bypassed operationally

Evidence-based reasoning	Outputs generated without provenance
Auditability	No immutable operational traceability
Confidence awareness	Confidence values absent or meaningless
Controlled execution	AI systems connected directly to infrastructure
Multi-domain validation	Cross-domain reasoning occurs without arbitration

Table 1. Governance requirements and common architectural failure modes.

2.6 Architectural Differentiation

Table 2 positions EAAP relative to five existing categories of AI deployment architecture and governance approach. The comparison is intended to clarify what EAAP provides that each existing category does not, rather than to evaluate those categories on their own terms.

Existing Approach	What It Lacks for Enterprise Governance	What EAAP Provides
Generic chatbot systems	Governance enforcement; no action tiers, approval boundaries, or audit trails	Layered governance with evidence classification, tiered approvals, and immutable audit records
Single-agent AI systems	Domain isolation; cross-domain reasoning without arbitration	Isolated domain agents coordinated through a governed orchestration layer
Orchestration-only frameworks (e.g. AutoGen, LangGraph)	Governance constraints; optimise for task completion, not auditability or controlled execution	Conditional Override, confidence gates, evidence classification, and execution boundaries
Policy-only governance (e.g. NIST AI RMF, ISO 42001)	Structural enforcement; governance principles remain advisory documents	Architectural control patterns that enforce policy as executable governance structure
Direct automation systems (RPA, infrastructure scripts)	AI reasoning layer, human approval model, and audit trail	Plan-Approve-Execute workflow with human approval gates and traceable execution
Model alignment frameworks (RLHF, Constitutional AI)	Operational system governance; focus on model behaviour rather than system architecture	System-level governance independent of the underlying model training approach

Table 2. EAAP architectural differentiation from existing approaches.

3. The Architectural Problem

The dominant architectural mistake in enterprise AI is treating the large language model as the system itself. This creates several structural problems: vendor lock-in, absent execution boundaries, opaque reasoning provenance, and governance bypass. EAAP positions the

model as a reasoning component operating within a governed architectural framework, separating reasoning, orchestration, governance, and execution as distinct architectural concerns.

3.1 Terminology: Governance Determinism

The term "deterministic" as used in this paper requires explicit definition. In formal computation, a deterministic system produces identical outputs for identical inputs. AI systems based on large language models are not deterministic in this sense: outputs vary due to sampling and non-deterministic inference. EAAP makes no claim of computational determinism for the underlying model.

"Governance determinism" is a new architectural term introduced by this paper within the EAAP framework. It is not derived from and is not intended to reference formal methods, computability theory, or any existing determinism concept in the computer science literature. Any apparent terminological similarity is incidental.

Definition: Governance Determinism

An AI system is governance-deterministic if: (a) the conditions governing whether a recommendation proceeds, escalates, or is blocked are defined in advance and applied uniformly; (b) evidence classification and confidence requirements for each action tier are structurally enforced rather than advisory; and (c) the execution pathway for any approved action follows a fixed, auditable control sequence independent of model output variation.

The model may produce varied reasoning outputs. The governance framework constrains which of those outputs may proceed to action, under what conditions, and with what human involvement. This is the operational meaning of governance determinism in EAAP.

4. Threat Model and Failure Modes

4.1 Overview

EAAP is a governance architecture. To establish what it governs against, this section defines the threat model: the set of failure modes that EAAP is explicitly designed to constrain. A governance architecture that does not define its threat model provides no meaningful assurance.

Nine threat categories are identified. Each represents a structural failure mode observable in current enterprise AI deployments. Table 3 provides a summary. Sections 4.2 and beyond describe each threat and the corresponding EAAP mitigation mechanism.

Threat	Description	EAAP Mitigation
Hallucinated Recommendations	Model produces plausible but incorrect recommendations without grounding in validated evidence	Mandatory evidence classification; RAG grounding in Knowledge and Evidence Layer; Assumption class forces flagging of unverified premises

Governance Bypass	AI outputs reach operational execution without completing required governance checks	Automation Gateway is the only authorised execution interface; GR-006 requires validated approval token; governance is enforced structurally, not through model instruction
Prompt Manipulation	Adversarial input causes the AI system to violate governance constraints or exceed authorised scope	GR-010 prohibits governance bypass through prompt manipulation; domain scope boundaries are architectural, not prompt-controlled; orchestration validates outputs against governance schema
Evidence Poisoning	The Knowledge and Evidence Layer is compromised with incorrect or adversarial reference material	Knowledge layer access controls; evidence source classification; Assumption class flagging; audit trail of evidence references used in each recommendation
Domain Contamination	Incorrect assumptions from one domain propagate into another, producing correlated errors across outputs	Domain isolation principle (GR-002); agents cannot directly alter other domain conclusions; all cross-domain interaction routes through the orchestration layer
Automation Bias	Human approvers over-trust AI recommendations, reducing approval to a rubber-stamp exercise	Confidence disclosure and evidence classification make recommendation basis explicit; Conditional Override forces approver engagement with uncertainty; tiered model reduces approver fatigue [4][8]
Stale Approvals	An approval issued under one set of conditions is used after conditions have materially changed	Approval expiry windows enforced in Section 7.3; validity prechecks at execution time; expired tokens rejected automatically by the Automation Gateway
Orchestration Deadlock	Multi-domain conflict resolution produces no resolvable output, halting the pipeline without surfacing the reason	GR-008 requires unresolved conflicts to trigger escalation; deadlock is a governance event requiring human review, not a silent failure condition
Execution Boundary Bypass	The AI system or an intermediary acquires direct infrastructure access, circumventing the Automation Gateway	GR-007 is a hard architectural rule; RBAC and allowlists constrain gateway scope; audit records detect unauthorised execution paths

Table 3. EAAP threat model: nine identified failure modes and corresponding mitigation mechanisms.

4.2 Threat Interdependencies

Several threats compound when they occur together. Evidence poisoning combined with governance bypass produces a scenario in which incorrect, externally sourced recommendations reach operational execution without human review. Domain contamination

combined with automation bias produces correlated errors across multiple infrastructure domains that approvers are statistically unlikely to detect. EAAP's layered architecture is specifically designed to contain compounding threats. No single failure can cascade through all layers without triggering a governance event. The Conditional Override mechanism, the approval tier model, and the Automation Gateway each represent independent containment boundaries.

5. EAAP Core Principles

EAAP is built around ten core architectural principles. Table 4 defines each principle. EAAP defines these principles independent of specific vendors, orchestration platforms, or underlying model providers. The domain agent examples used throughout this paper are illustrative. Any compliant implementation using different tooling, models, or infrastructure platforms remains architecturally valid.

Principle	Description
Governance First	Governance is structural and mandatory. Controls are enforced architecturally, not through policy alone.
Deterministic Control	Operational behaviour is governed by pre-defined, uniformly applied rules per the definition in Section 3.1.
Domain Isolation	Agents reason only within bounded technical domains. Cross-domain interaction is mediated by the orchestration layer.
Human Accountability	Humans remain operationally and legally responsible for all outcomes.
Evidence Transparency	All outputs require evidence classification disclosing provenance and confidence basis before proceeding.
Confidence Awareness	Uncertainty must be explicitly represented in operationally meaningful terms.
Sovereign Deployment	External dependencies are minimised to support air-gapped and data-sovereign operation.
Execution Separation	AI systems must not directly execute infrastructure actions. The Automation Gateway is the only execution interface.
Auditability	All actions and recommendations must be traceable and immutably recorded.
Conditional Override	Insufficient confidence, incomplete evidence, or unresolved conflicts trigger escalation rather than autonomous continuation.

Table 4. EAAP core architectural principles.

6. EAAP Architecture Overview

6.1 The Layered Model

EAAP is organised as a layered architecture model with six distinct layers, each with clearly defined responsibilities and execution boundaries. Table 5 summarises layer functions. Figure 1 presents the layered architecture model.

Layer	Function
Governance Layer	Defines operational constraints and approval requirements
Orchestration Layer	Coordinates reasoning and arbitration across domains
Domain Intelligence Layer	Performs specialised domain reasoning
Knowledge and Evidence Layer	Supplies validated reference material
Execution Layer	Interfaces with operational systems through gateways
Audit and Traceability Layer	Maintains immutable operational records

Table 5. EAAP architectural layers and functions.

Figure 1. EAAP Layered Architecture Model



Figure 1 - EAAP layered architecture model.

Upper layers govern lower layers. The Execution Layer interfaces exclusively with operational infrastructure through the Automation Gateway.

6.2 Domain Isolation

Domain agents are isolated reasoning units. Each agent reasons only within its own domain, consumes domain-appropriate evidence, and cannot directly alter another domain's conclusions. Cross-domain interaction occurs only through the orchestration layer. This prevents domain contamination, cascading reasoning failures, and uncontrolled assumption propagation, directly addressing the domain contamination threat in Section 4.

6.3 Governance-Deterministic Orchestration

EAAP orchestration follows a structured lifecycle. Each phase is mandatory and sequenced. Figure 2 presents the orchestration lifecycle.

Figure 2. EAAP Governance-Deterministic Orchestration Lifecycle

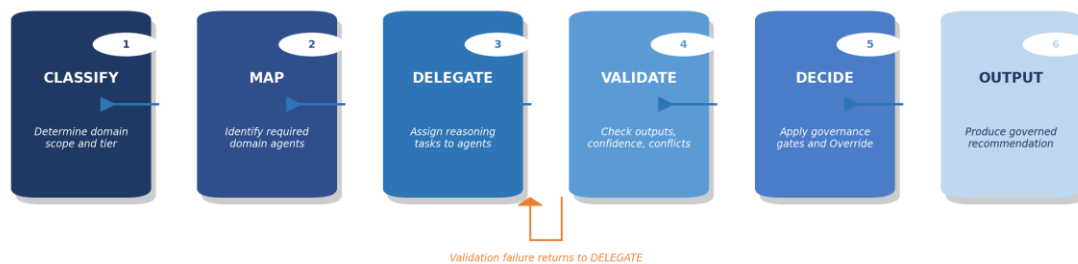


Figure 2 - EAAP orchestration lifecycle.

Each phase must complete before the next begins. Validation failure returns to DELEGATE.

The orchestration layer identifies required domains, delegates reasoning tasks, validates outputs, detects conflicts, and synthesises final recommendations. It does not replace domain reasoning. It governs it.

7. The Governance Layer

7.1 Evidence Classification

All EAAP outputs are classified using five evidence categories. Classification is mandatory and cannot be omitted. Table 6 defines the evidence classes. This directly addresses the hallucinated recommendations and evidence poisoning threats identified in Section 4.

Evidence Class	Description
Vendor-Supported	Official vendor documentation, validated architectures, or vendor-published guidance
Framework-Supported	Recognised industry standards or governance frameworks (ISO, NIST, BSI)
Best Practice	Industry-accepted operational guidance without formal standardisation
Heuristic	Expert operational reasoning based on practitioner experience

Assumption	Explicitly unverified premise that must be flagged and requires operator validation before execution
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Table 6. EAAP evidence classification model.

7.2 Confidence and Uncertainty

EAAP treats uncertainty as operationally significant, consistent with the interpretability requirements established by Doshi-Velez and Kim [16]. Every recommendation must disclose known unknowns, expose uncertainty explicitly, and identify confidence limitations. Where confidence falls below required thresholds, recommendations are blocked, escalated, or routed for additional validation through the Conditional Override model.

7.3 Human Approval Boundaries

Human approval is structurally enforced. High-impact operational actions require explicit approval, change references, approval expiry windows, and auditable authorisation chains. Table 7 defines the four operational tiers.

Tier	Classification	Example	Approval Requirement
Tier 0	Read-only	Status queries, reporting	None required
Tier 1	Low-risk reversible	Non-disruptive configuration reads	Governance layer validation
Tier 2	Disruptive operational	VM migrations, storage reconfigurations	Mandatory human approval
Tier 3	High-impact or destructive	Data deletion, security policy changes	Mandatory human approval with change record

Table 7. EAAP operational action tier model.

8. The Orchestration Layer

8.1 Supervisory Coordination

The orchestration layer acts as a supervisory governance engine. Responsibilities include domain selection, orchestration sequencing, dependency awareness, confidence aggregation, and governance enforcement. It is the only component permitted to merge reasoning outputs from multiple domains.

8.2 Conflict Resolution

When domain outputs conflict, the orchestration layer identifies contradictions, evaluates evidence quality, and determines whether sufficient confidence exists to continue. Unresolved conflicts trigger escalation. All detected conflicts must be surfaced in the output regardless of

whether resolution is achieved. Hidden conflict suppression is prohibited by GR-008, directly addressing the orchestration deadlock threat.

8.3 Conditional Override Logic

Conditional Override is triggered when confidence is insufficient, evidence is incomplete, governance requirements are unmet, or domain conflicts remain unresolved. The system does not produce authoritative recommendations in these cases. It exposes missing inputs and escalates for human review. This mechanism prevents false certainty, addressing automation bias risk documented by Parasuraman and Manzey [4] and Cummings [8].

9. The Domain Intelligence Layer

Domain agents operate as specialist reasoning systems. Each agent has bounded operational scope, consumes domain-specific knowledge, follows structured output schemas, and cannot autonomously execute operational changes. Grounding is provided by the Knowledge and Evidence Layer through RAG pipelines [11], allowing agents to retrieve validated reference material before generating outputs rather than relying on model weights alone.

Examples of domain agents include virtualisation agents (VMware vSphere, Nutanix AHV), storage governance agents, cybersecurity analysis agents, compliance review agents (ISO 27001, BSI IT-Grundschutz, EU AI Act), network configuration agents, and identity management agents. These examples are illustrative. EAAP does not prescribe specific vendors or platforms for any domain agent role.

The infrastructure examples used throughout this paper are illustrative and do not limit EAAP applicability to infrastructure operations or operational technology environments.

10. The Execution Layer

10.1 Automation Gateway Model

EAAP prohibits direct AI access to operational infrastructure. All operational interaction occurs through an Automation Gateway. The gateway exposes controlled API surfaces, implements RBAC, enforces allowlists, validates approvals, and records audit data. The AI system never directly executes infrastructure actions. This implements a hard execution boundary addressing the execution boundary bypass threat and consistent with NIST Zero Trust Architecture principles [13].

10.2 Plan-Approve-Execute Workflow

Operational execution follows a four-stage workflow. Figure 3 presents the workflow model.

Figure 3. EAAP Plan-Approve-Execute-Audit Workflow

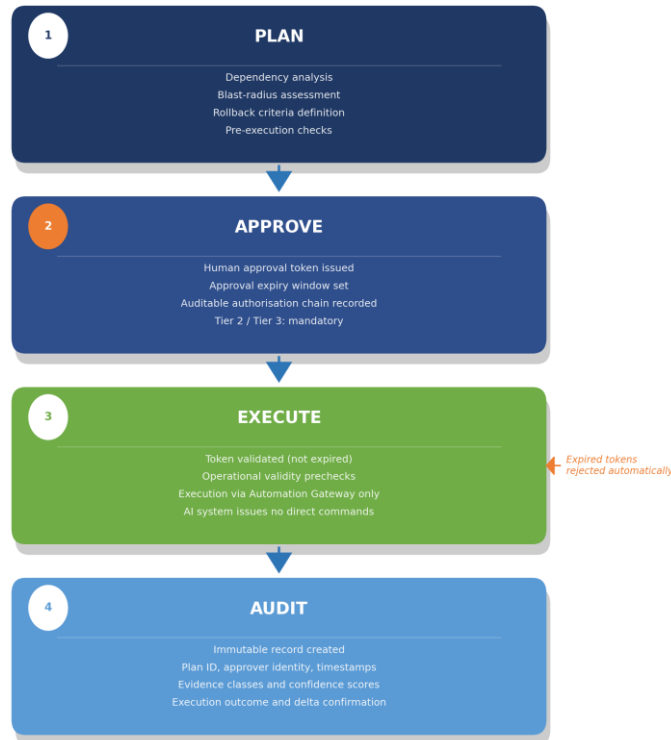


Figure 3 -EAAP Plan-Approve-Execute-Audit workflow.

Expired or invalid approval tokens are rejected automatically at the Execution stage.

The planning stage includes dependency analysis, blast-radius assessment, rollback criteria, and prechecks. Execution requires valid approval tokens and operational validity checks. Expired or invalid approvals are rejected automatically, directly addressing the stale approvals threat.

11. Air-Gapped and Sovereign Deployment

EAAP is designed to support sovereign and air-gapped deployment models. The framework assumes environments where external APIs may be prohibited, Internet connectivity may not exist, and operational data sovereignty is mandatory. This includes defence environments, regulated industries, critical infrastructure, and national-security-adjacent operations.

The framework supports local model inference using open-weight models, local vector storage for RAG pipelines, locally hosted orchestration, and offline knowledge packs. All EAAP layers are designed to operate without external connectivity, reducing data exfiltration risk and external dependency exposure.

12. Auditability and Traceability

EAAP requires operational traceability across recommendations, approvals, evidence references, orchestration decisions, and execution outcomes. Audit records include timestamps, user identity, plan identifiers, approval references, confidence classifications, evidence classes used, and execution metadata.

The framework assumes immutable audit retention for operationally significant actions, consistent with ISO/IEC 42001:2023 [3] and the NIST AI RMF [2] accountability requirements. Audit records must be retained in a tamper-evident form accessible to governance reviewers independently of the AI system that produced them.

13. Illustrative Scenario

13.1 Scenario Overview

The following scenario demonstrates how EAAP governs a realistic Tier 2 operational action in a Nutanix AHV enterprise environment. An operator submits a request to migrate 24 virtual machines from a source cluster to a target cluster during a scheduled maintenance window. The migration involves storage reconfiguration and network path changes affecting production workloads.

13.2 Governance Lifecycle Trace

Step 1: Request Classification (Orchestration Layer)

The orchestration layer classifies the request as a multi-domain operational change affecting storage, network, and compute domains. Tier assignment: Tier 2 (disruptive operational action). Required domains: virtualisation, storage, networking.

Step 2: Domain Reasoning (Domain Intelligence Layer)

Three domain agents are invoked in parallel. Virtualisation agent: Vendor-Supported evidence (Nutanix AHV live migration documentation). Storage agent: Vendor-Supported for capacity checks, Assumption for target performance under concurrent load. Networking agent: Best Practice (no explicit vendor guidance for this topology).

Step 3: Confidence Assessment (Governance Layer)

Combined confidence: 74 percent. Tier 2 threshold: 85 percent. Conditional Override invoked. The storage Assumption is unvalidated and the networking evidence is below the Vendor-Supported threshold required for Tier 2 actions.

Step 4: Escalation (Conditional Override)

The system surfaces the confidence shortfall, identifies the specific missing evidence, blocks the execution pathway, and routes the case to the operator for additional input. No authoritative recommendation is produced.

Step 5: Evidence Supplementation

The operator provides storage performance benchmarks and a confirmed network topology diagram. Revised combined confidence: 91 percent. Conditional Override is cleared.

Step 6: Human Approval (Governance Layer)

Tier 2 mandatory approval workflow triggered. An approval request is issued with the plan identifier, evidence summary, blast-radius assessment, and rollback criteria. The approver issues an approval token with a 4-hour expiry window.

Step 7: Execution (Automation Gateway)

The approval token is validated. Prechecks confirm host capacity, replication status, and network availability. Execution proceeds through the Automation Gateway. The AI system issues no direct infrastructure commands.

Step 8: Audit Record

An immutable audit record is created containing the request identifier, timestamps, operator identity, domain outputs with evidence classes, confidence scores, Conditional Override event log, approval reference, and execution outcome.

13.3 Scenario Observations

Three core EAAP behaviours are demonstrated. First, the system identified its own confidence shortfall without producing a false-certainty recommendation: governance failure was surfaced, not suppressed. Second, human approval was structurally required, not advisory: the execution pathway was blocked until a valid approval token was issued. Third, the large language model served as a reasoning component within a governed pipeline and did not exercise operational authority at any stage.

Step	Layer	Governance Action
1. Classification	Orchestration	Tier 2 assigned; domains identified
2. Domain reasoning	Domain Intelligence	Parallel reasoning; evidence classified
3. Confidence assessment	Governance	74% < 85% threshold; Conditional Override triggered
4. Escalation	Orchestration	Missing evidence surfaced; execution blocked
5. Evidence supplementation	Governance	Revised confidence 91%; Override cleared
6. Human approval	Governance	Approval token issued with expiry window
7. Execution	Execution	Token validated; Automation Gateway executes
8. Audit	Audit and Traceability	Immutable record created

Table 8. EAAP governance lifecycle trace for the illustrative Tier 2 scenario.

14. What EAAP Cannot Do and Non-Goals

14.1 What EAAP Cannot Do

EAAP is intended to constrain AI systems, not grant them autonomous authority. Table 9 defines the framework limitations.

Limitation	Explanation
Cannot guarantee correctness	AI recommendations remain probabilistic regardless of governance controls
Cannot eliminate human error	Humans remain operationally responsible; EAAP cannot prevent approver error
Cannot resolve incomplete evidence	Missing evidence remains a governance problem; EAAP surfaces but cannot fill gaps
Cannot fully prevent automation bias	Operators may still over-trust AI outputs despite disclosure mechanisms [4][8]
Cannot replace enterprise governance	Organisational governance, policy, and compliance remain separately necessary
Cannot remove operational risk	Governance reduces but does not eliminate risk
Cannot make autonomous execution safe	Execution boundaries are mandatory and cannot be compensated by other controls

Table 9. EAAP framework limitations.

14.2 Non-Goals

The following items are explicit non-goals of EAAP. They are stated to prevent category errors in architectural interpretation and to reduce misdirected criticism.

EAAP is not:

- An AGI framework or general autonomous reasoning system. The framework is explicitly designed to constrain, not enable, AI autonomy.
- An autonomous execution framework. Human approval remains mandatory for Tier 2 and Tier 3 actions. The framework does not provide a pathway for unattended operational execution.
- A guarantee of operational correctness. EAAP governs the process by which AI recommendations are generated and authorised; it does not guarantee that those recommendations are correct.
- A replacement for enterprise governance, compliance, or risk management functions. EAAP supplements these functions architecturally; it does not supersede them.
- A model alignment framework. EAAP does not specify training methodology, reward functions, or model-level safety techniques. It operates above the model layer.
- A substitute for human accountability and professional responsibility. GR-007 is explicit: AI systems advise, humans decide, gateways execute.
- A vendor-specific product, platform, or software implementation. EAAP is an architectural control pattern. No specific vendor, framework, or model is mandated.

15. Governing Rules

EAAP defines ten governing rules. These are non-negotiable architectural requirements, not implementation recommendations. Table 10 enumerates all rules.

Rule ID	Governing Rule
GR-001	Governance controls are structural and mandatory
GR-002	Domain agents operate only within assigned scope
GR-003	Evidence classification is mandatory for all recommendations
GR-004	Human approval is mandatory for Tier 2 and Tier 3 actions
GR-005	Known unknowns must be disclosed explicitly in all outputs
GR-006	Execution requires a validated, non-expired approval token
GR-007	AI systems advise; humans decide; gateways execute
GR-008	Unresolved domain conflicts trigger escalation, not suppression
GR-009	Auditability and traceability are mandatory for all significant actions
GR-010	Governance requirements cannot be bypassed through prompt manipulation

Table 10. EAAP governing rules.

16. Current Limitations

This section identifies acknowledged limitations of the EAAP framework in its current form. These are not future research goals but present constraints that bound the claims made in this paper.

Limitation	Description
No empirical validation	EAAP has not been evaluated in a controlled enterprise deployment. All claims are based on architectural reasoning, not measured outcomes.
Confidence thresholds undefined	The framework defines a confidence gate mechanism but does not specify how threshold values should be calibrated for different domain types or action tiers.
Orchestration conflict resolution	The protocol for resolving multi-domain conflicts is defined at the principle level only. No formal arbitration algorithm is specified.
Automation bias not fully addressed	Evidence disclosure reduces but does not eliminate automation bias. Human factors research is required to quantify the residual risk.
Audit schema undefined	The framework requires immutable audit records but does not define a standardised schema, retention policy, or interchange format.
Air-gapped model lifecycle	Secure update, validation, and rollback pipelines for models in isolated environments are outside the current scope of EAAP.

Table 11. Current EAAP framework limitations.

17. Future Research Directions

This section identifies research directions that would extend, validate, or formalise EAAP beyond its current scope. These are open problems, not acknowledged deficiencies of the current framework.

Research Direction	Description
Empirical validation	Controlled evaluation of EAAP governance outcomes across multiple enterprise deployment environments and domain types
Formal specification	Mathematical formalisation of governance determinism per the approach of Seshia et al. [18], enabling proof-based assurance of governance lifecycle correctness
Confidence calibration methodology	Development of a principled method for setting and validating confidence thresholds across domain types, evidence classes, and action tiers
Human factors research	Empirical studies of automation bias in EAAP-governed environments to quantify the residual risk of over-trust despite evidence disclosure
Governance interoperability	Integration of EAAP governance records with enterprise ITSM, SIEM, and CMDB tooling through standardised interchange formats
Cross-domain reasoning assurance	Formal assurance methods for orchestration correctness in adversarial scenarios, including evidence poisoning and prompt manipulation
EAAP standardisation	Engagement with standards bodies (ISO/IEC JTC 1/SC 42, NIST) to assess alignment and potential standardisation of EAAP governance principles

Table 12. Future research directions for EAAP development and formalisation.

EAAP is intended to evolve through future publications as empirical evidence, formal analysis, and implementation experience accumulate. The author invites technical collaboration from enterprise architects, AI governance researchers, operational engineering teams, and standards communities.

18. Conclusion

Enterprise AI systems increasingly operate in environments where operational trust, auditability, governance, and controlled execution matter more than raw model capability. This paper has argued that the primary challenge in enterprise AI is architectural rather than purely algorithmic.

EAAP proposes a governance-oriented architecture pattern that structures AI reasoning, constrains operational execution, isolates domain intelligence, exposes uncertainty, enforces human approval, and maintains operational auditability. The framework positions the large language model as a replaceable reasoning component within a larger governed system. A formal threat model defines nine failure modes the architecture is designed to address.

The framework is implementation-neutral: EAAP defines governance principles independent of specific vendors, models, or orchestration platforms. Governance determinism, as defined in this paper, is an architectural property of the system as a whole, not a property of any individual component.

Current limitations are acknowledged. Empirical validation, confidence calibration, and formal specification remain open research areas. The governance gap in enterprise AI is real and growing. Addressing it requires architectural commitment, not merely policy intent.

Central Principle
AI systems may assist operational reasoning, but governance authority, accountability, and execution responsibility remain structurally human.

Appendix A: Terminology

This appendix defines the primary architectural terms introduced or given specific meaning within the EAAP framework. These definitions are binding for the purposes of interpreting this paper and any subsequent EAAP publications.

Term	Definition
Governance Determinism	A new architectural term introduced by this paper. An AI system is governance-deterministic if its governance conditions are defined in advance, uniformly applied, and structurally enforced, independent of model output variation. Distinct from computational determinism in formal methods.
Conditional Override	A governance mechanism triggered when confidence is insufficient, evidence is incomplete, or domain conflicts are unresolved. Prevents false-certainty recommendations by blocking the execution pathway and escalating to human review.
Domain Isolation	The architectural principle that each domain agent reasons only within its assigned technical domain. Cross-domain interaction is mediated exclusively by the orchestration layer. Prevents domain contamination and assumption propagation.
Confidence Gate	A governance control that compares aggregated domain confidence against a pre-defined tier threshold. Recommendations that do not meet the threshold trigger Conditional Override rather than proceeding to the approval stage.
Evidence Classification	The mandatory assignment of an evidence class (Vendor-Supported, Framework-Supported, Best Practice, Heuristic, or Assumption) to all AI outputs, disclosing the provenance and reliability basis of each recommendation.
Sovereign AI	AI system deployment in which all components, including model inference, knowledge storage, and orchestration, operate within a defined data sovereignty boundary without external network dependencies. Applicable to air-gapped and regulated environments.
Execution Isolation	The architectural requirement that AI systems interact with operational infrastructure only through the Automation Gateway. Direct execution by the AI system or its intermediary components is prohibited.

Automation Gateway	The sole authorised interface between the EAAP execution layer and operational infrastructure. Implements RBAC, allowlist enforcement, approval token validation, and execution audit logging.
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Table A1. EAAP terminology definitions.

References

- [1] Floridi, L. et al. (2018). AI4People — An Ethical Framework for a Good AI Society. *Minds and Machines*, 28(4), 689–707. <https://www.research-collection.ethz.ch/server/api/core/bitstreams/47a754b6-979f-46c6-9c98-9485e8ffdc1/content>
- [2] NIST. (2023). Artificial Intelligence Risk Management Framework (AI RMF 1.0). National Institute of Standards and Technology. <https://www.nist.gov/itl/ai-risk-management-framework>
- [3] ISO/IEC 42001:2023. Artificial Intelligence Management Systems. <https://www.iso.org/standard/81230.html>
- [4] Parasuraman, R., & Manzey, D. H. (2010). Complacency and Bias in Human Use of Automation. *Human Factors*, 52(3), 381–410. https://www.researchgate.net/publication/47792928_Complacency_and_Bias_in_Human_Use_of_Automation_An_Attentional_Integration
- [5] United Nations. (2023). Governing AI for Humanity. Advisory Body on Artificial Intelligence. <https://digitallibrary.un.org/record/4062495?v=pdf>
- [6] European Union. (2024). EU Artificial Intelligence Act. <https://eur-lex.europa.eu/eli/reg/2024/1689/oj>
- [7] Scharre, P. (2018). *Army of None: Autonomous Weapons and the Future of War*. W.W. Norton & Company. <https://www.jstor.org/stable/pdf/26891891.pdf>
- [8] Cummings, M. L. (2004). Automation Bias in Intelligent Time Critical Decision Support Systems. AIAA Intelligent Systems Technical Conference. <https://arc.aiaa.org/doi/10.2514/6.2004-6313>
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This paper is a conceptual governance architecture proposal. It has not undergone formal peer review. The author welcomes technical discussion and critical review from enterprise architects, AI governance researchers, operational engineering teams, and standards communities.